Supplementary Problems — Vector Potential, Self-Inductance, Mutual Inductance

Selected problems from Nayfeh & Brussel

This document complements HW5 with eight further problems drawn from Nayfeh & Brussel's *Electricity and Magnetism*, chosen to hammer three closely related topics: the **magnetic vector potential A**, **self-inductance** L, and **mutual inductance** M. The three are tightly linked — both inductances are most cleanly computed from a flux, and a flux is most cleanly computed from the line integral of $\mathbf{A}$ around the loop (Stokes's theorem, $\Phi = \oint \mathbf{A} \cdot d\mathbf{l}$). I have tried, as in HW5, to make the physical reasoning visible alongside the algebra, not hide it.

Notation: $\mathbf{A}$ is the vector potential (satisfying $\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}$), $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H/m}$, cylindrical radial coordinate is ρ (Nayfeh's convention), and "per unit length" inductances are written L/l.

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Problem 8.7 — Vector potential of two antiparallel line currents

Setup. Two infinite parallel line currents carry $+I$ and $-I$ in the $\hat{\mathbf{a}}_z$ direction. Let ρ_1 and ρ_2 be the perpendicular distances from an observation point to the $+I$ and $-I$ wires, respectively. We are to show

$$\mathbf{A} \;=\; \hat{\mathbf{a}}_z \,\frac{\mu_0 I}{2\pi} \ln\!\frac{\rho_2}{\rho_1}.$$

Strategy

Because $\mathbf{A}$ satisfies the same Poisson-type equation as the scalar potential does for charges (just one component at a time, driven by the current density), and because the problem is linear, we can simply *superpose* the vector potentials of two independent wires.

A for a single infinite wire

For an infinite wire along the z-axis carrying current I in $+\hat{\mathbf{a}}_z$, the field is $\mathbf{B} = \hat{\mathbf{a}}_\phi\, \mu_0 I/(2\pi\rho)$. Seeking $\mathbf{A} = \hat{\mathbf{a}}_z A_z(\rho)$ and using $(\boldsymbol{\nabla}\times\mathbf{A})_\phi = -\frac{\partial A_z}{\partial \rho}$,

$$-\frac{\partial A_z}{\partial \rho} = \frac{\mu_0 I}{2\pi\rho} \quad\Longrightarrow\quad A_z(\rho) = -\frac{\mu_0 I}{2\pi}\ln\rho + C.$$

The additive constant is a choice of gauge/zero-reference and does not affect any physical quantity.

Superpose

Let wire 1 (current $+I$) produce $A_z^{(1)} = -(\mu_0 I/2\pi)\ln\rho_1$ and wire 2 (current $-I$) produce $A_z^{(2)} = +(\mu_0 I/2\pi)\ln\rho_2$. Adding:

$$\boxed{\mathbf{A}(\mathbf{r}) \;=\; \hat{\mathbf{a}}_z\,\frac{\mu_0 I}{2\pi}(\ln\rho_2 - \ln\rho_1) \;=\; \hat{\mathbf{a}}_z\,\frac{\mu_0 I}{2\pi}\ln\frac{\rho_2}{\rho_1}.}$$

The constants of integration from each wire combine into one overall constant that we absorbed (equivalently, we fixed the gauge so that $\mathbf{A}\to 0$ on the perpendicular bisector where $\rho_1 = \rho_2$).

Sanity check. On the bisector ($\rho_1 = \rho_2$), $\mathbf{A} = 0$ by symmetry: the two contributions exactly cancel there. Near the $+I$ wire, $\rho_1 \to 0$ and $\mathbf{A} \to -\infty$ along $\hat{\mathbf{a}}_z$; near the $-I$ wire, $\rho_2 \to 0$ and $\mathbf{A} \to +\infty$ along $\hat{\mathbf{a}}_z$. Both are correct: A_z diverges logarithmically at a filamentary source, with the *sign* of the divergence matching the sign of the current. ✓

Problem 8.8 — Three vector potentials for a uniform field

Setup. For a uniform magnetic field $\mathbf{B} = B\,\hat{\mathbf{a}}_z$, three candidate vector potentials are offered:

$$\mathbf{A}_1 = -By\,\hat{\mathbf{a}}_x, \qquad \mathbf{A}_2 = Bx\,\hat{\mathbf{a}}_y, \qquad \mathbf{A}_3 = -\tfrac{1}{2}\,\mathbf{r}\times\mathbf{B}.$$

We verify each is a valid vector potential, compute each divergence, show $\mathbf{A}_1 - \mathbf{A}_2$ is a gradient, and sketch the fields.

Verifying $\boldsymbol{\nabla}\times\mathbf{A}_i = \mathbf{B}$

The only nonzero component of $\boldsymbol{\nabla}\times\mathbf{A}$ we need is z. In Cartesian: $(\boldsymbol{\nabla}\times\mathbf{A})_z = \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}$.

For $\mathbf{A}_1 = -By\,\hat{\mathbf{a}}_x$: $A_x = -By$, $A_y = 0$, so $(\boldsymbol{\nabla}\times\mathbf{A}_1)_z = 0 - (-B) = B$. ✓

For $\mathbf{A}_2 = Bx\,\hat{\mathbf{a}}_y$: $A_x = 0$, $A_y = Bx$, so $(\boldsymbol{\nabla}\times\mathbf{A}_2)_z = B - 0 = B$. ✓

For $\mathbf{A}_3$: Expand with $\mathbf{r} = x\hat{\mathbf{a}}_x + y\hat{\mathbf{a}}_y + z\hat{\mathbf{a}}_z$ and $\mathbf{B} = B\hat{\mathbf{a}}_z$:

$$\mathbf{r}\times\mathbf{B} = B\,(x\hat{\mathbf{a}}_x + y\hat{\mathbf{a}}_y + z\hat{\mathbf{a}}_z)\times\hat{\mathbf{a}}_z = B\,[x(\hat{\mathbf{a}}_x\times\hat{\mathbf{a}}_z) + y(\hat{\mathbf{a}}_y\times\hat{\mathbf{a}}_z)] = B\,(-x\hat{\mathbf{a}}_y + y\hat{\mathbf{a}}_x).$$

So $\mathbf{A}_3 = -\frac{1}{2}\mathbf{r}\times\mathbf{B} = \frac{B}{2}(x\hat{\mathbf{a}}_y - y\hat{\mathbf{a}}_x)$. Then

$$(\boldsymbol{\nabla}\times\mathbf{A}_3)_z = \frac{\partial}{\partial x}\Big(\tfrac{Bx}{2}\Big) - \frac{\partial}{\partial y}\Big(-\tfrac{By}{2}\Big) = \tfrac{B}{2} + \tfrac{B}{2} = B. \quad ✓$$

Divergences

Using $\boldsymbol{\nabla}\cdot\mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$:

$$\boldsymbol{\nabla}\cdot\mathbf{A}_1 = \frac{\partial(-By)}{\partial x} = 0, \quad \boldsymbol{\nabla}\cdot\mathbf{A}_2 = \frac{\partial(Bx)}{\partial y} = 0, \quad \boldsymbol{\nabla}\cdot\mathbf{A}_3 = -\tfrac{B}{2}\frac{\partial y}{\partial x} + \tfrac{B}{2}\frac{\partial x}{\partial y} = 0.$$

All three potentials satisfy $\boldsymbol{\nabla}\cdot\mathbf{A} = 0$ — they are all in Coulomb gauge.

$\mathbf{A}_1 - \mathbf{A}_2$ as a gradient

$$\mathbf{A}_1 - \mathbf{A}_2 \;=\; -By\,\hat{\mathbf{a}}_x \;-\; Bx\,\hat{\mathbf{a}}_y.$$

We claim this equals $\boldsymbol{\nabla}\psi$ for some ψ. Try $\psi = -Bxy$:

$$\boldsymbol{\nabla}(-Bxy) = -By\,\hat{\mathbf{a}}_x - Bx\,\hat{\mathbf{a}}_y. \quad \checkmark$$

So $\mathbf{A}_1 - \mathbf{A}_2 = \boldsymbol{\nabla}(-Bxy)$, which confirms the general principle: any two vector potentials for the same $\mathbf{B}$ differ by the gradient of a scalar function — this is **gauge freedom**. Since the curl of a gradient vanishes, adding $\boldsymbol{\nabla}\psi$ to $\mathbf{A}$ doesn't change $\mathbf{B}$.

Sketches in the xy-plane

- $\mathbf{A}_1 = -By\,\hat{\mathbf{a}}_x$**:** horizontal arrows, magnitude growing linearly with $|y|$, pointing in $-\hat{\mathbf{a}}_x$ for $y > 0$ and $+\hat{\mathbf{a}}_x$ for $y < 0$.
- $\mathbf{A}_2 = Bx\,\hat{\mathbf{a}}_y$**:** vertical arrows, magnitude growing linearly with $|x|$, pointing in $+\hat{\mathbf{a}}_y$ for $x > 0$ and $-\hat{\mathbf{a}}_y$ for $x < 0$.
- $\mathbf{A}_3 = (B/2)(x\hat{\mathbf{a}}_y - y\hat{\mathbf{a}}_x)$**:** circulating arrows around the origin, magnitude growing linearly with $\sqrt{x^2+y^2}$ — this is the most symmetric choice.

They all produce the same $\mathbf{B} = B\hat{\mathbf{a}}_z$; they just "hide" it in different looking patterns of $\mathbf{A}$.

Problem 8.11 — Vector potential of a long wire from $\boldsymbol{\nabla}\times\mathbf{A} = \mathbf{B}$

Setup. Determine $\mathbf{A}$ for an infinite filamentary wire along $\hat{\mathbf{a}}_z$ carrying current I, working directly from $\boldsymbol{\nabla}\times\mathbf{A} = \mathbf{B}$ and Ampère's law.

Step 1 — B from Ampère

Azimuthal symmetry $\Rightarrow \mathbf{B} = \hat{\mathbf{a}}_\phi B_\phi(\rho)$. An Amperian loop of radius ρ gives

$$\oint \mathbf{H}\cdot d\mathbf{l} = H_\phi \cdot 2\pi\rho = I \quad\Longrightarrow\quad \mathbf{B} = \hat{\mathbf{a}}_\phi\,\frac{\mu_0 I}{2\pi\rho}.$$

Step 2 — Guess the form of A

The source $\mathbf{J}$ points in $\hat{\mathbf{a}}_z$, and $\mathbf{A}$ inherits the direction of the current (from the Biot–Savart-like form $\mathbf{A} = (\mu_0/4\pi)\int \mathbf{J}/R\,dv'$). Translational symmetry along z kills z-dependence; azimuthal symmetry kills ϕ-dependence. So

$$\mathbf{A} = \hat{\mathbf{a}}_z\,A_z(\rho).$$

Step 3 — Apply $\nabla \times \mathbf{A} = \mathbf{B}$

In cylindrical coordinates, for $\mathbf{A} = \hat{\mathbf{a}}_z A_z(\rho)$,

$$\nabla \times \mathbf{A} = -\hat{\mathbf{a}}_\phi \frac{\partial A_z}{\partial \rho}.$$

Matching to $\mathbf{B} = \hat{\mathbf{a}}_\phi \, \mu_0 I/(2\pi\rho)$:

$$-\frac{\partial A_z}{\partial \rho} = \frac{\mu_0 I}{2\pi\rho} \quad \Longrightarrow \quad A_z(\rho) = -\frac{\mu_0 I}{2\pi} \ln \rho + C.$$

Step 4 — Fix the gauge

Choosing the zero reference at some radius ρ_0 (the wire surface, or any convenient reference — infinity does *not* work here because A_z diverges logarithmically):

$$\boxed{\mathbf{A}(\rho) \;=\; \hat{\mathbf{a}}_z \frac{\mu_0 I}{2\pi} \ln\frac{\rho_0}{\rho}.}$$

Remark on infinity. The logarithmic divergence is a genuine feature of infinite straight currents (much as the 2D electrostatic potential of an infinite line charge diverges logarithmically). For a *finite* wire, $\mathbf{A}$ is well-defined everywhere and $\to 0$ at infinity; the logarithmic blow-up only appears after we take the infinite-length idealization.

Problem 11.10 — Self-inductance of a square-cross-section toroid

Setup. A toroid of inner radius ρ_1, outer radius ρ_2, and axial thickness a (square cross section $\Leftrightarrow a = \rho_2 - \rho_1$ though that is not assumed in what follows), with N closely-wound turns carrying current I.

Step 1 — B inside the toroid

Apply Ampère's law to a circular loop of radius ρ (with $\rho_1 < \rho < \rho_2$) centered on the toroid axis. The loop links N turns, each carrying I, so

$$H_\phi \cdot 2\pi\rho = NI \quad \Longrightarrow \quad \mathbf{B} = \hat{\mathbf{a}}_\phi \frac{\mu_0 NI}{2\pi\rho}.$$

Outside the toroid's core, $\mathbf{B} = 0$ (the opposing contributions of the inner and outer portions of each winding cancel, for an Amperian loop that either encloses no turns or passes through each turn twice with opposite orientation).

Step 2 — Flux through one turn

The cross-section is a rectangle of width $\rho_2 - \rho_1$ (radial) and height a (axial). $\mathbf{B}$ threads the cross-section in $\hat{\mathbf{a}}_\phi$, which is normal to it. A horizontal strip at radius ρ of width $d\rho$ has area $a\,d\rho$:

$$\Phi_1 \;=\; \int_{\rho_1}^{\rho_2} \frac{\mu_0 NI}{2\pi\rho} \, a \, d\rho \;=\; \frac{\mu_0 NIa}{2\pi} \ln\frac{\rho_2}{\rho_1}.$$

Step 3 — Total flux linkage and inductance

Each of the N turns links the same Φ_1, so the flux linkage is $\lambda = N\Phi_1$ and

$$\boxed{L = \frac{\lambda}{I} = \frac{\mu_0 N^2 a}{2\pi} \ln\frac{\rho_2}{\rho_1}.}$$

Check against the "solenoid" limit. If $\rho_2 - \rho_1 \ll \rho_1$ (thin toroid), let $\rho_2 = \rho_1 + w$ with $w \ll \rho_1$. Then $\ln(\rho_2/\rho_1) \approx w/\rho_1$, so

$$L \approx \frac{\mu_0 N^2 a w}{2\pi\rho_1} = \frac{\mu_0 N^2 A}{\ell}, \qquad A = aw, \;\; \ell = 2\pi\rho_1,$$

recovering the straight-solenoid formula $L = \mu_0 N^2 A/\ell$ with cross-sectional area A and winding length ℓ. ✓

Problem 11.12 — Self-inductance of a coaxial line

Setup. Coaxial cylindrical shells of inner radius a and outer radius b, with permeability μ in the annulus. Current I flows along the inner conductor and returns on the outer shell. Find L/l (neglecting the internal inductance of the conductor metal itself — the shells are assumed thin).

B in the annulus

For $a < \rho < b$, an Amperian loop of radius ρ encloses the full current I, giving

$$\mathbf{B} = \hat{\mathbf{a}}_\phi \frac{\mu I}{2\pi\rho}.$$

Outside the outer shell ($\rho > b$), net enclosed current is zero, so $\mathbf{B} = 0$; inside the inner conductor ($\rho < a$) we are treating the conductor as thin and ignoring its internal field (this is why we are computing only the "external" inductance).

Flux per unit length

Consider a rectangular surface of axial length l bounded by the inner and outer conductors. Flux through it:

$$\Phi = l\int_a^b \frac{\mu I}{2\pi\rho}\, d\rho = \frac{\mu I l}{2\pi} \ln\frac{b}{a}.$$

Since the return current is on the outer shell, the cable constitutes a single-turn "loop" per length, so the flux linkage is just Φ itself, and

$$\boxed{\frac{L}{l} = \frac{\mu}{2\pi} \ln\frac{b}{a}.}$$

Comment on the internal inductance. A solid inner conductor of radius a carrying uniformly distributed current contributes an additional internal inductance of $\mu_{\text{conductor}}/(8\pi)$ per unit length (derivable from the magnetic energy stored in the $\rho < a$ region). For high-frequency applications where the skin effect confines the current to the conductor surface, this internal term vanishes and the formula above is exact.

Problem 11.15 — Parallel two-wire transmission line

Setup. Two identical parallel conductors of radius a, separation d (center-to-center), carrying equal and opposite currents I.

Field between the conductors

Place wire 1 at the origin (current $+I$ in $\hat{\mathbf{a}}_z$) and wire 2 at $x = d$ (current $-I$). On the line segment connecting their centers, both contributions add (a filament at x sees wire 1 at distance x pushing $\mathbf{B}$ one way, and wire 2 at distance $d - x$ pushing $\mathbf{B}$ the *same* way because its current is reversed):

$$\mathbf{B}(x) \;=\; \hat{\mathbf{a}}_\phi \,\frac{\mu_0 I}{2\pi}\left[\frac{1}{x} + \frac{1}{d-x}\right], \qquad a < x < d - a.$$

Flux per unit length

Take a rectangular strip of axial length l spanning $x \in [a,\, d-a]$ (from the surface of wire 1 to the surface of wire 2):

$$\Phi = l\int_a^{d-a} \frac{\mu_0 I}{2\pi}\left[\frac{1}{x} + \frac{1}{d-x}\right] dx = \frac{\mu_0 I l}{2\pi}\left[\ln\frac{d-a}{a} + \ln\frac{d-a}{a}\right] = \frac{\mu_0 I l}{\pi}\,\ln\frac{d-a}{a}.$$

Inductance per unit length

$$\boxed{\frac{L}{l} \;=\; \frac{\mu_0}{\pi}\,\ln\frac{d-a}{a} \;\approx\; \frac{\mu_0}{\pi}\,\ln\frac{d}{a} \quad (d \gg a).}$$

Connection to the $\cosh^{-1}$ form

The exact external inductance (accounting properly for the fact that each wire is a *cylindrical* conductor of finite radius, not a filament, using bipolar coordinates) can be written

$$\frac{L}{l} \;=\; \frac{\mu_0}{\pi}\,\cosh^{-1}\frac{d}{2a}.$$

Using the identity $\cosh^{-1}(u) = \ln\!\left(u + \sqrt{u^2 - 1}\right)$ with $u = d/(2a)$:

$$\cosh^{-1}\frac{d}{2a} = \ln\left(\frac{d}{2a} + \sqrt{\frac{d^2}{4a^2} - 1}\right).$$

For $d \gg 2a$, the square root $\approx d/(2a)$, so $\cosh^{-1}(d/2a) \approx \ln(d/a)$ — matching our elementary formula above.

Numerical evaluation (part b)

For $d = 25\,\text{ft}$ and $a = 0.8\,\text{in}$, convert to common units:

$$d = 25 \cdot 12\ \text{in} = 300\ \text{in}, \qquad \frac{d}{2a} = \frac{300}{1.6} = 187.5.$$

Then $\cosh^{-1}(187.5) \approx \ln(2 \cdot 187.5) = \ln(375) \approx 5.927$. So

$$\frac{L}{l} \;\approx\; \frac{4\pi \times 10^{-7}}{\pi} \cdot 5.927 \;\approx\; \boxed{2.37\ \mu\text{H/m}.}$$

Equivalently about $0.72\ \mu$H/ft — a typical value quoted for open-wire overhead transmission lines.

Problem 11.5 — Rectangle sliding alongside a long wire

Setup. A long straight wire carries steady current I in $\hat{\mathbf{a}}_z$. A coplanar rectangular loop of dimensions a (width, transverse to the wire) by b (length, parallel to the wire) is at perpendicular distance l from its near long side. The loop moves with speed v perpendicular to the wire ("parallel to its width").

Step 1 — Mutual inductance

At perpendicular distance x from the wire, $\mathbf{B} = \hat{\mathbf{a}}_\phi \, \mu_0 I/(2\pi x)$, normal to the loop's plane. A thin strip of width dx at distance x has area $b\,dx$. Integrating across the width of the loop:

$$\Phi \;=\; \int_l^{l+a} \frac{\mu_0 I}{2\pi x}\, b\, dx \;=\; \frac{\mu_0 I b}{2\pi}\, \ln\!\frac{l+a}{l}.$$

Hence

$$\boxed{M(l) \;=\; \frac{\Phi}{I} \;=\; \frac{\mu_0 b}{2\pi}\, \ln\!\left(1+\frac{a}{l}\right).}$$

Notice M depends on the instantaneous position l — this is essential for part (a).

Step 2 — Induced emf

As the loop moves, $l = l(t)$ with $dl/dt = v$ (positive if it recedes from the wire). The flux through the loop is $\Phi(l) = M(l)\, I$, and Faraday's law gives

$$\varepsilon = -\frac{\mathrm{d}\Phi}{\mathrm{d}t} = -I\,\frac{\mathrm{d}M}{\mathrm{d}l}\cdot\frac{\mathrm{d}l}{\mathrm{d}t} = -Iv\,\frac{\mathrm{d}M}{\mathrm{d}l}.$$

Compute the derivative:

$$\frac{\mathrm{d}M}{\mathrm{d}l} = \frac{\mu_0 b}{2\pi}\,\frac{\mathrm{d}}{\mathrm{d}l}\ln\!\frac{l+a}{l} = \frac{\mu_0 b}{2\pi}\left[\frac{1}{l+a}-\frac{1}{l}\right] = -\frac{\mu_0 b a}{2\pi l(l+a)}.$$

Therefore

$$\boxed{\varepsilon(l) \;=\; \frac{\mu_0 I b a v}{2\pi\, l(l+a)}.}$$

Sign and intuition. As the loop *recedes* ($v > 0$), the flux decreases (M decreases with l), so by Lenz's law the induced emf drives a current that *maintains* the flux — the induced current in the loop flows in the same sense as the source current's image would suggest. The magnitude scales as $1/l^2$ for $l \gg a$, consistent with the intuition that distant loops feel weak fields.

Problem 11.8 — Neumann formula: two fixed wires and a moving wire

Setup. Two infinitely long fixed wires sit at $x = \pm a$, parallel to the y-axis. They carry currents $\mp I$ (so the wire at $x = +a$ carries $-I\,\hat{\mathbf{a}}_y$ and the wire at $x = -a$ carries $+I\,\hat{\mathbf{a}}_y$). A third, infinitely long parallel wire moves in the plane of the other two; its instantaneous distances from the wires at $x = +a$ and $x = -a$ are r_1 and r_2 respectively.

Part (a) — Mutual inductance per unit length

Neumann's formula for the mutual inductance between two circuits is

$$M = \frac{\mu_0}{4\pi} \oint\oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{|\mathbf{r}_1 - \mathbf{r}_2|}.$$

For parallel infinite filaments this gives a divergent integral, but the *per-unit-length* mutual inductance is well-defined through the flux or, equivalently, through the vector potential.

The fixed two-wire system is precisely the configuration of Problem 8.7, so its vector potential is

$$\mathbf{A}(\mathbf{r}) \;=\; \hat{\mathbf{a}}_y \, \frac{\mu_0 I}{2\pi} \, \ln\frac{r_2}{r_1},$$

where r_1, r_2 are distances to the $+a$ and $-a$ wires respectively. For the moving wire (parallel to $\hat{\mathbf{a}}_y$), the flux linkage per unit length is simply A_y evaluated at its location:

$$\Phi \text{ per unit length} = A_y \cdot 1 = \frac{\mu_0 I}{2\pi} \ln\frac{r_2}{r_1}.$$

Therefore

$$\boxed{\frac{M}{l} \;=\; \frac{\mu_0}{2\pi} \ln\frac{r_2}{r_1}.}$$

Why this works: $\Phi = \oint \mathbf{A} \cdot d\mathbf{l}$ around the moving circuit. For an infinite wire returning at infinity, only the straight-segment piece survives per unit length, giving A_y times that length.

Part (b) — Induced emf when the moving wire is at the origin

At the instant the moving wire passes through the origin ($x = 0$), it is equidistant from both fixed wires: $r_1 = r_2 = a$, so $\ln(r_2/r_1) = 0$ — the mutual inductance *itself* is zero there. But the emf depends on $\dot{M}$, not M, so we need

$$\frac{\mathrm{d}M}{\mathrm{d}t} = \frac{\mathrm{d}M}{\mathrm{d}x} \cdot \frac{\mathrm{d}x}{\mathrm{d}t} = v_0 \frac{\mathrm{d}M}{\mathrm{d}x}.$$

Write $r_1 = |x - a|$, $r_2 = |x + a|$. For the wire passing through the origin with $-a < x < a$, both absolute values open with positive arguments, giving $r_1 = a - x$ and $r_2 = a + x$:

$$\frac{M}{l}(x) = \frac{\mu_0}{2\pi}\big[\ln(a + x) - \ln(a - x)\big].$$

Differentiate:

$$\frac{\mathrm{d}(M/l)}{\mathrm{d}x} = \frac{\mu_0}{2\pi}\left[\frac{1}{a + x} + \frac{1}{a - x}\right] = \frac{\mu_0}{2\pi} \cdot \frac{2a}{a^2 - x^2}.$$

At $x = 0$: $\frac{\mathrm{d}(M/l)}{\mathrm{d}x}\big|_{x=0} = \mu_0/(\pi a)$.

The emf per unit length in the moving wire is

$$\varepsilon \;=\; -I \, \frac{\mathrm{d}(M/l)}{\mathrm{d}t} \;=\; -I\, v_0 \, \frac{\mathrm{d}(M/l)}{\mathrm{d}x}\bigg|_{x=0} \;=\; -\frac{\mu_0 I v_0}{\pi a}.$$

$$\boxed{|\varepsilon| \;=\; \frac{\mu_0 I v_0}{\pi a} \text{ per unit length.}}$$

Motional-emf cross-check. At the origin, the moving wire sees the combined field of the two fixed wires:

$$\mathbf{B}(0) = \hat{\mathbf{a}}_z \frac{\mu_0 I}{2\pi a} + \hat{\mathbf{a}}_z \frac{\mu_0 I}{2\pi a} = \hat{\mathbf{a}}_z \frac{\mu_0 I}{\pi a}.$$

(The contributions add because the currents at $\pm a$ are antiparallel, so their $\mathbf{B}$-fields point the same way at the origin.) The motional emf per unit length is $|\mathbf{v} \times \mathbf{B}| \cdot 1 = v_0 \cdot \mu_0 I/(\pi a)$, which agrees exactly. ✓

Summary & methodological takeaways

Across these eight problems three themes recur.

Theme 1 — Vector potential as a computational lever. Problems 8.7, 8.11, and 11.8 all use $\mathbf{A}$ not as a mathematical curiosity but as a concrete shortcut: superposition is trivial, and the line-integral form $\Phi = \oint \mathbf{A} \cdot d\mathbf{l}$ converts a surface integral over an awkward region into a loop integral, often picking up only one or two parallel segments. Problem 8.8 is the reminder that $\mathbf{A}$ is only defined up to a gradient (gauge freedom) — three very different-looking potentials can produce the same $\mathbf{B}$.

Theme 2 — Inductance = flux linkage per unit current, done carefully. For self-inductance (11.10, 11.12, 11.15) the recipe is uniform: use symmetry + Ampère to get $\mathbf{B}$, integrate across the cross-section to get Φ, multiply by the number of turns if the geometry is wound, and divide by I. The subtlety is keeping track of *external* vs. *internal* contributions (the latter coming from field inside a finite-thickness conductor).

Theme 3 — Mutual inductance is a geometric quantity. In Problem 11.5 the mutual inductance $M(l)$ is a function of geometry alone — the current I in the source wire never appears in the formula for M. The induced emf shows up only when M *changes* in time, whether by motion (11.5, 11.8) or by a time-varying current (not treated here). Neumann's formula and the $\Phi = \oint \mathbf{A} \cdot d\mathbf{l}$ approach are two faces of the same calculation, and for systems of parallel wires the second is usually faster.

These three techniques — vector potential, flux linkage, and Neumann/$\mathbf{A}$-line-integral — cover the overwhelming majority of inductance calculations one encounters in practice. When the geometry is too complicated for any of them, energy methods ($U = \frac{1}{2}LI^2 = \frac{1}{2\mu_0}\int B^2 dv$) are the next tool to reach for.